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To cite this article: C H L Peters *et al* 2011 *Physiol. Meas.* **32** 1517

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Physiol. Meas. **32** (2011) 1517–1527 doi:10.1088/0967-3334/32/10/001

A continuous wavelet transform-based method for time-frequency analysis of artefact-corrected heart rate variability data

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Received 23 January 2011, accepted for publication 27 July 2011
Published 18 August 2011
Online at stacks.iop.org/PM/32/1517

Abstract

Time-frequency analysis of heart rate variability (HRV) provides relevant clinical information. However, time-frequency analysis is very sensitive to artefacts. Artefacts that are present in heart rate recordings may be corrected, but this reduces the variability in the signal and therefore adversely affects the accuracy of calculated spectral estimates. To overcome this limitation of traditional techniques for time-frequency analysis, a new continuous wavelet transform (CWT)-based method was developed in which parts of the scalogram that have been affected by artefact correction are excluded from power calculations. The method was evaluated by simulating artefact correction on HRV data that were originally free of artefacts. Commonly used spectral HRV parameters were calculated by the developed method and by the short-time Fourier transform (STFT), which was used as a reference. Except for the powers in the very low-frequency and low-frequency (LF) bands, powers calculated by the STFT proved to be extremely sensitive to artefact correction. The CWT-based calculations in the high-frequency and very high-frequency bands corresponded well with their theoretical values. The standard deviations of these powers, however, increase with the number of corrected artefacts which is the result of the non-stationarity of the R–R interval series that were analysed. The powers calculated in the LF band turned out to be slightly sensitive to artefact correction, but the results were acceptable up to 20% artefact correction. Therefore, the CWT-based method provides a valuable

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alternative for the analysis of HRV data that cannot be guaranteed to be free of artefacts.

Keywords: continuous wavelet transform, heart rate variability, artefacts

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Together with stroke volume and peripheral vascular resistance, heart rate is one of the main cardiovascular parameters that is regulated by the autonomic nervous system to control the supply of blood to the organs. To fulfil this task, the cardiovascular control centres in the medulla oblongata receive input from various types of sensors in the body. Two very important types of sensor are the pressure-sensitive baroreceptors, which are located in the walls of the arteries and in the heart, and the chemoreceptors, which are located in the carotid arteries, the aortic arch and the medulla oblongata. In response to this sensory input, the heart rate is controlled through both neurological and hormonal pathways. Consequently, changes in the heart rate reflect the sensory input to the cardiovascular control centres. Therefore, changes in the heart rate indirectly reflect changes in the physiological condition. However, due to the complexity of the cardiovascular regulation, it is difficult to obtain unambiguous information on changes in the physiological condition from a recording of the heart rate (Notarius and Floras 2001, Parati *et al* 2006).

To a certain extent, the hormonal and neurological pathways that are used to control the heart rate operate on different time bases. More in particular, the sympathetic and parasympathetic parts of the autonomic nervous system act on very characteristic time scales. Time-frequency analysis of heart rate variability (HRV) can therefore be used to evaluate sympathetic and parasympathetic nervous activity in regulating the heart rate (Akselrod *et al* 1981). Currently, time-frequency analysis of HRV is applied widely in clinical research in various fields of medicine (Mainardi *et al* 2002, Acharya *et al* 2006). Successful applications in clinical practice have, however, remained scarce. This is partially due to the complexity of the cardiovascular control mechanisms. Additionally, for accurate results the beat-to-beat heart rate is required, and also data must be preprocessed adequately. Finally, time-frequency analysis is very sensitive to artefacts, and therefore an artefact-free heart rate signal is required (Clifford and Tarassenko 2005, Peters *et al* 2008).

In clinical practice, artefact-free signals cannot always be guaranteed. An illustrative example is the non-invasive measurement of the foetal ECG from the abdomen of a pregnant woman. Time-frequency analysis of foetal HRV can provide valuable information on foetal condition (van Laar *et al* 2008, Siira *et al* 2005, Rantonen *et al* 2001, Cerutti *et al* 1989, van Laar *et al* 2010). However, antepartum the quality of the measured ECG signals can be rather poor, as the electrical source of the signals—the foetal heart—is still very small and only accessible through maternal tissue (Lewis 2003). Foetal heart rates obtained from the maternal abdomen are therefore often contaminated by artefacts.

The most frequently occurring artefacts in beat-to-beat heart rate signals result from

(i) incorrect localization of the QRS complex (misdetected heart beat),
(ii) one or multiple undetected QRS complexes (missed heart beats) and
(iii) false detection of a QRS complex (phantom heart beat).

Each of these conditions results in artefacts in the calculated heart rate, which must be detected and corrected before performing time-frequency analysis. Artefact correction involves removing phantom heart beats from the recording, replacing misdetected heart beats and inserting missed heart beats in the recording. For replacement and insertion, substitute QRS locations are usually obtained by means of interpolation, autoregressive modelling or other statistical methods. Irrespective of the method that is used to replace and insert QRS locations, artefact correction will reduce the variability in the signal. Therefore, it is inevitable that artefact correction will adversely affect the accuracy of calculated spectral estimates. This limits the application of traditional techniques for time-frequency analysis to data that are almost entirely free of artefacts.

To overcome this severe limitation, a new method for time-frequency analysis of HRV is introduced. The method that is described in this work is based on the continuous wavelet transform (CWT) and can be applied for time-frequency analysis, after artefact correction, of HRV data that originally contained considerable amounts of artefacts. However, this work does not address the actual detection of these artefacts, which may be a challenge itself, nor does it address the process of artefact correction for which various methods exist. Instead, artefact correction was simulated on HRV data that were originally free of artefacts. This approach allows objective evaluation of the developed method for various amounts of artefacts, but implicitly assumes adequate artefact detection and correction. As a reference, the generated data are also analysed by the short-time Fourier transform (STFT).

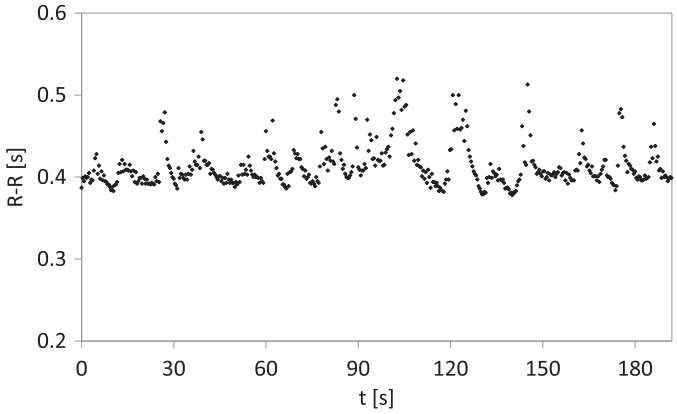
2. Methodology

2.1. General approach

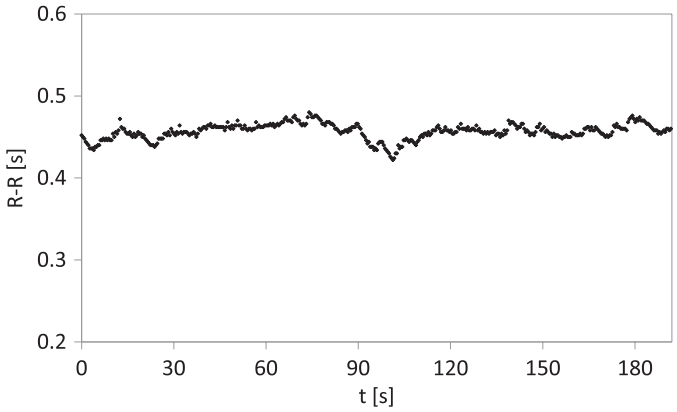
Beat-to-beat heart rate data from newborns and foetuses were used to study the effect of artefact correction on time-frequency analysis of HRV. In newborns, the respiration rate is significantly higher than in adults. As the heart rate is modulated by respiration, neonatal heart rate data will contain variability at frequencies of up to 1.5 Hz. Foetal heart rate data will also contain variability at high frequencies, mainly due to respiratory movements. In adult heart rate data, variability is usually absent in this frequency range. Using neonatal and foetal heart rate data therefore provides a unique opportunity to include the effect of artefact correction on high-frequency spectral estimates into the evaluation.

(i) *Neonatal data.* Neonatal heart rate data were obtained in the neonatal intensive care unit (NICU) of the Máxima Medical Center. Three-lead ECG signals from a General Electric Solar 8000M patient monitor were recorded at a sample rate of 240 Hz. In these ECG signals, QRS complexes were detected using a quadratic R-fit to reduce sampling errors. This resulted in a dataset containing time stamps for the R-peak in all QRS complexes that were detected in the recorded signal. Five segments of 192 s that were completely free of artefacts were selected from this dataset for further analysis. Figure 1(a) shows the R–R interval series for one of the five segments.

(ii) *Foetal data.* Foetal heart rate data were obtained in the obstetrics department of the Máxima Medical Center. During delivery, single-lead direct foetal scalp ECG signals were recorded from a Neoventa STAN s31 foetal monitor at a sample rate of 500 Hz. A dataset with time stamps for the R-peak in all QRS complexes that were detected by the foetal monitor was obtained from the digital output of the STAN s31. Five segments of 192 s that were completely free of artefacts were selected from this dataset for further analysis. Figure 1(b) shows the R–R interval series for one of the five segments.



(a) Neonatal heart rate data



(b) Foetal heart rate data

Figure 1. Examples of artefact-free neonatal (a) and foetal (b) heart rate data.

To simulate artefacts, timestamps were deleted at random positions in the foetal and neonatal data segments. Timestamps were deleted until the desired number of artefacts was achieved. Next, R–R intervals were calculated from these data, using linear interpolation to substitute data that were marked as the artefact. This way, the R–R interval series were created that simulated artefact-corrected heart rate data at ten different numbers of artefacts, varying from 5% to 50% of the segment length in equal steps. For each number of artefacts, each of the foetal and neonatal data segments was reused to generate 20 different datasets from each original segment. This resulted in a total of 2000 different data segments that were analysed by the developed CWT-based method and by the STFT.

2.2. New CWT-based method

The standard technique for frequency analysis is the Fourier transform. When applied for time-frequency analysis, the Fourier transform is characterized by its constant resolution in time and frequency, which is fixed by the choice of sample rate and window length. Consequently, the time-frequency plane of the spectrogram is tiled in Heisenberg boxes of constant dimensions. For the analysis of HRV in this study, this is a major disadvantage as the presence of corrected

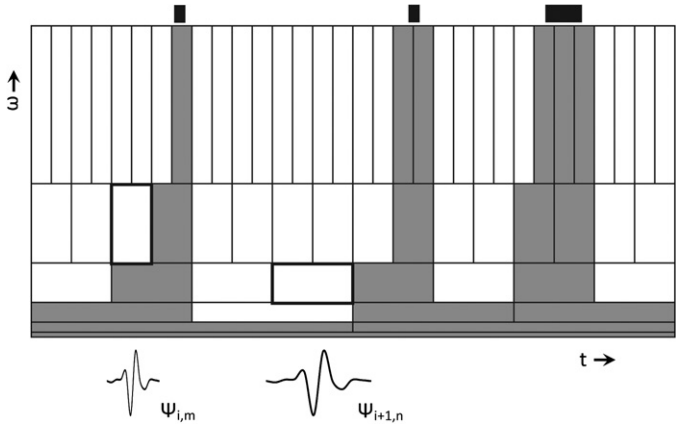


Figure 2. Tiling of the time-frequency plane for a dyadic wavelet decomposition. Artefact locations are indicated above the figure in black, and the affected time-frequency tiles are coloured grey. The wavelet functions $\psi_{i,m}$ and $\psi_{i+1,n}$ correspond to the bold outlined Heisenberg boxes.

artefacts will influence the calculated spectrogram for the entire window length. However, artefact correction will mainly affect the higher frequencies in the spectrogram. Therefore, a shorter time resolution in these frequencies would limit the duration of the influence of artefact correction and allow for discarding parts of the spectrogram that are affected by artefact correction.

A technique that allows for multi-resolution analysis is the wavelet transform, which decomposes the input signal by convolution with scaled and dilated functions of a mother wavelet function (Mallat 1998, Addison 2002, Akansu and Haddad 2001). An important natural feature of wavelet transforms is that time and frequency resolution are scalable, with the restriction that the area of the Heisenberg boxes must remain constant. Figure 2 shows the tiling of the scalogram for a traditional dyadic wavelet decomposition. The figure also illustrates the duration of the effect that artefact correction has for different wavelet scales.

Because of its scalable time and frequency resolution, the wavelet transform qualifies for analysis of non-stationary signals and therefore has been successfully applied for the analysis of HRV (Wiklund *et al* 1997, Kimura *et al* 1998, Pichot *et al* 1999). A dyadic decomposition is however far from ideal for the analysis, as the specific frequency bands of interest (Task force of the European Society of Cardiology and the North American Society of Pacing Electrophysiology 1996) will not be matched exactly by dyadic scales. Consequently, the focus of most studies that used the wavelet transform for the analysis of HRV has been on matching these frequency bands. Elegant solutions were found in the use of wavelet packets (Tanaka and Hargens 2004) and in integrating the scalogram of the CWT (David *et al* 2007).

Table 1 contains an overview of the spectral parameters of interest and their corresponding frequency bands. The very low-frequency (VLF) band reflects the long-term regulation of the heart rate, including hormonal and thermoregulation. The low-frequency (LF) band is associated with both the sympathetic and the parasympathetic branches of the autonomic nervous system, whereas higher frequencies mainly reflect parasympathetic nervous activity. As vagal nervous activity is modulated by respiration, the main contribution to the power at high frequencies is found at the respiration frequency. For adults, the high-frequency (HF) band is commonly defined as 0.15–0.4 Hz. Due to a higher respiration frequency, this frequency band is defined as 0.4–1.5 Hz (VHF) for foetuses and newborns. It is common

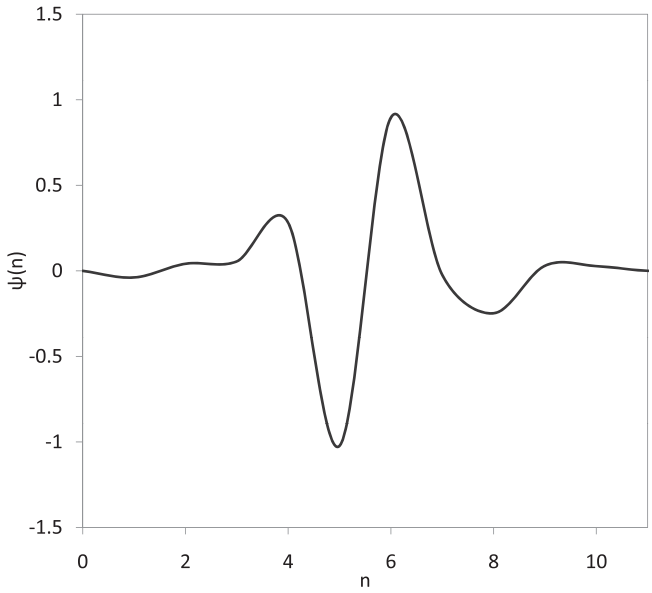


Figure 3. Fifth-order symlet mother wavelet function.

Table 1. Overview of spectral parameters of interest.

Symbol	Description	Frequency band
VLF	Power in very low-frequency band	$<$ 0.04 Hz
LF	Power in low-frequency band	0.04–0.15 Hz
HF	Power in high-frequency band	0.15–0.4 Hz
VHF	Power in very high-frequency band (foetus/newborn)	0.4–1.5 Hz
TP	Total power (outside VLF)	0.04–1.5 Hz
LFn	LF power in normalized units	
VHFn	VHF power in normalized units	
LF/VHF	LF/VHF ratio	

to express the powers in the frequency bands of interest in normalized units (LFn, VHFn) as these parameters may more reliably characterize autonomic cardiovascular control. These normalized LF and VHF powers are calculated by dividing the powers in, respectively, the LF and VHF bands by the total power (TP) between 0.04 and 1.5 Hz.

In our approach, the time resolution of a CWT is exploited to discard parts of the scalogram that are affected by artefact correction. A selection of specific scales from a CWT is used to match the frequency bands of interest. Based on the additional requirements of orthogonality and a short support width, the fifth-order symlet wavelet was selected for the analysis of HRV. Figure 3 displays the mother wavelet function of this wavelet. A least-squares calculation was used to select the scales of the CWT that correspond with the frequency bands of interest.

Due to their beat-to-beat character, the R–R interval series are not equidistantly distributed in time. Before performing the CWT, the R–R interval series were transformed into an equidistant set of data. This was achieved by applying a sample and hold technique to the R–R interval series. The resulting signal was convolved with a square function of width

0.5 s and energy equal to 1, and then resampled at a rate of 4 Hz (Peters *et al* 2004, de Jong 2000). This way, an equidistant dataset was obtained, in which each data point corresponds to the weighted average of the R–R intervals in the 0.5 s surrounding the sample. The CWT was calculated from scale 0.1 to scale 80 in steps of 0.1. To correct for the convolution with the square function during the resampling procedure, the calculated wavelet coefficients are divided by the coefficients of the wavelet transform of this square function. Then, the power scalogram was calculated according to

$$P_{W_f}[n, s] = \frac{1}{C_g} \left| \frac{W_f[n, s]}{s^2} \right|. \tag{1}$$

In equation (1), $W_f[n, s]$ is the corrected wavelet coefficients, s is the scale and C_g is the admissibility constant of the mother wavelet function. The powers in the frequency bands of interest can be calculated from the power scalogram by time averaging over the duration of the segment and integrating over scale, as given by

$$P_{xF} = \sum_{s \in xF} \frac{1}{T} \sum_{n} P_{W_f}[n, s] \Delta t \Delta s. \tag{2}$$

To reduce the effect of corrected artefacts for powers in the HF bands, the matrix $A[n, s]$ is introduced to eliminate specific parts of the scalogram from the calculation. The power in the HF band is then calculated by

$$P_{HF} = \sum_{s \in HF} \frac{\sum_n (A[n, s] \circ P_{W_f}[n, s])}{\sum_n A[n, s]} \Delta t \Delta s. \tag{3}$$

In equation (3), $(A[n, s] \circ P_{W_f}[n, s])$ denotes the Hadamard product, or entrywise product, of matrices $A[n, s]$ and $P_{W_f}[n, s]$, whose elements are given by

$$(A[n, s] \circ P_{W_f}[n, s])_{i,j} = (A[n, s])_{i,j} \cdot (P_{W_f}[n, s])_{i,j}. \tag{4}$$

For artefact-free parts of the R–R interval series, the elements of matrix $A[n, s]$ are equal to 1. For the VHF frequency band, $A[n, s]$ is equal to zero around locations of corrected artefacts for one-third of the support width of scale s, which was considered an acceptable effective support width. A longer effective support width would exclude unnecessary large parts of the R–R interval series from the analysis, while not significantly contributing to further reduction of the effect of artefact correction. For the HF frequency band, the number elements that are set to zero around corrected artefacts were limited to one-third of the support width of the smallest scale in that frequency band, as it is expected that the effect of artefact correction on calculated powers will be smaller for larger scales, and longer effective support widths are undesirable. For the LF and VLF frequency bands, $A[n, s]$ is equal to 1 for all scales for the entire R–R interval series. By pointwise multiplying $A[n, s]$ with the scalogram, parts of the scalogram that are strongly affected by artefact correction are discarded in the calculation of the powers in the two HF bands.

2.3. Reference method

As a reference method for time-frequency analysis of the generated R–R interval series, the STFT, or windowed Fourier transform, was used. Identical to the preprocessing for the CWT, the R–R interval series were transformed into an equidistant set of data before processing. The STFT was calculated using a 256-point Parzen window and a window shift of 0.25 s. Before application of the Parzen window, the DC offset of the R–R intervals within the window was removed to prevent spectral leakage. The calculated Fourier transforms are squared to obtain the power spectrogram, and then corrected for the energy reduction due to the application

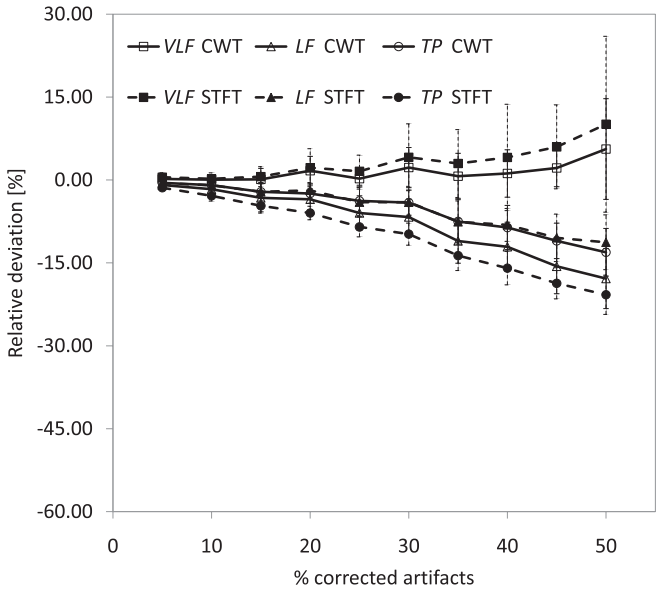


Figure 4. Relative deviations in VLF, LF and TP and their standard deviations for both the CWT-based method and the STFT.

of the Parzen window (Peters *et al* 2004, de Jong 2000). This correction is made by dividing the spectrogram by the ratio of the power in the Parzen window and the power in a square window of identical length. Also the spectrogram is corrected for the convolution of the input signal with the square function by means of division by a squared sinc function, the Fourier transform of the square function that was used (Peters *et al* 2004, de Jong 2000).

3. Results

All 2000 segments of foetal and neonatal R–R interval series with various numbers of corrected artefacts were analysed using both the CWT-based method and the STFT. For all calculated spectral estimates, the relative deviation from their theoretical value—the value calculated for the artefact-free data segments—was determined. For both methods, figure 4 shows the relative deviations in VLF power, LF power and TP for all analysed segments. Figure 5 shows the relative deviations in the HF and VHF powers. Finally, figure 6 shows the relative deviations in the LFn and VHFn.

4. Discussion

4.1. Short-time Fourier transform

For the STFT analysis of HRV, the calculated powers in the LF and VLF range correspond well with their theoretical values for datasets in which up to 30% of the R–R interval series has been replaced by interpolated values. The powers in the HF and VHF ranges calculated by STFT however are significantly smaller than their theoretical values. This deviation already occurs at relative low numbers of corrected artefacts and increases rapidly as this number increases. This is a direct consequence of the reduced variability in these frequency ranges

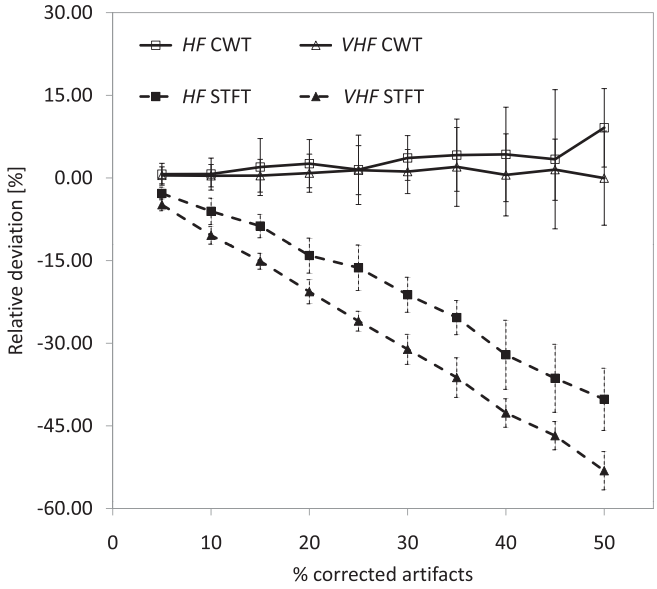


Figure 5. Relative deviations in HF and VHF and their standard deviations for both the CWT-based method and the STFT.

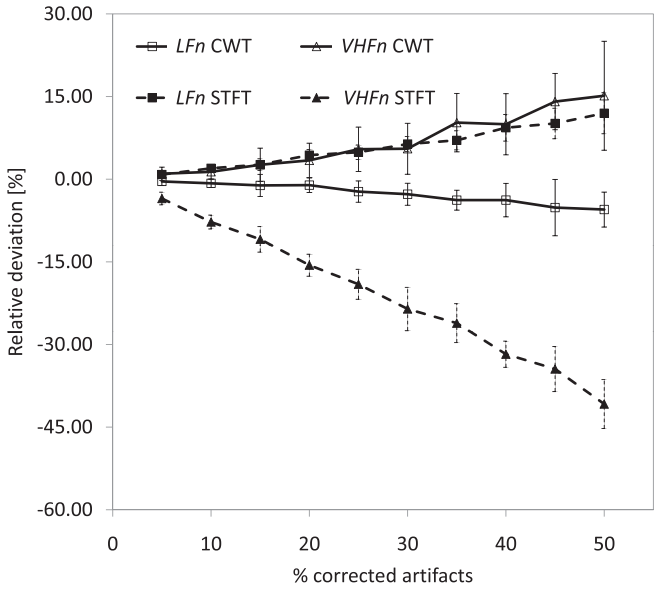


Figure 6. Relative deviations in LFn and VHFn and their standard deviations for both the CWT-based method and the STFT.

due to the replacement of artefacts by interpolated values. Although the deviations in the HF band are smaller than in the VHF band, in both ranges STFT analysis will not provide reliable parameters when the R–R interval series that are analysed originally contained artefacts.

Consequently, most other spectral parameters that cover these frequency ranges, such as TP, VHFn and LF/VHF ratio, can also not be calculated reliably for datasets that originally

contained artefacts. However, for the analysed datasets, the normalized LF power (LFn) could be calculated with acceptable accuracy as long as less than 25% of the dataset was replaced by interpolated values to correct for artefacts. This can be explained by the relatively very large amount of power that is present in the LF range, compared to higher frequencies. For the datasets that were analysed, this is quite common; however, for datasets in which this is not the case, it is expected that LFn powers will also be affected by artefact correction.

4.2. CWT-based method

Similar to the STFT analysis, the results in the VLF domain for the CWT-based method remain unaffected even for very high numbers of corrected artefacts. The calculated powers in the LF band on the other hand show an underestimation of their theoretical value, which grows with the number of corrected artefacts, but is acceptable when up to 20% of the dataset has been replaced with interpolated values. This effect is due to the trailing edge of the fifth-order symlet's frequency response in the HF range. The CWT powers in both the HF and VHF bands remain within a few per cent of their theoretical values even when up to 50% of the dataset has been replaced with interpolated values. The standard deviation of the calculated powers in these frequency bands increases with the number of corrected artefacts. Most likely, this is the result of the non-stationarity of the R–R interval series that are analysed. The calculated powers in the HF range will therefore fluctuate slightly, dependent on the exact location of artefacts in the original datasets.

Due to the relatively large amount of power in the LF range, the calculated TPs show almost similar underestimations as the calculated LF powers and therefore the normalized LF power is only slightly underestimated. The LF/VHF ratio is underestimated, which is the direct consequence of the underestimation of the LF power. Logically, VHFn overestimates its theoretical value. Nevertheless, also for these parameters, the correspondence with their theoretical values is acceptable when up to 20% of the dataset has been replaced with interpolated values.

5. Conclusion

Time-frequency analysis of HRV by means of the STFT is sensitive to artefact correction. Although LF parameters can be calculated accurately when up to 30% of an R–R interval series is replaced by interpolated values, it is not possible to calculate HF parameters reliably if artefacts were originally present in the dataset. The STFT analysis is therefore not suitable for time-frequency analysis of heart rate data that are not guaranteed to be free of artefacts.

By developing a CWT-based method for time-frequency analysis of HRV, parts of R–R interval series that originally contained artefacts can be discarded in the calculation of HF spectral estimates. By using this approach, HF spectral parameters can be calculated reliably even when extremely large parts of data are missing due to artefacts. However, compared to STFT, the calculation of spectral parameters in the LF range is more sensitive to artefacts but acceptable for datasets in which up to 20% of the data have been replaced with interpolated values to correct for artefacts.

The developed CWT-based method offers a valuable alternative for time-frequency analysis of HRV data that contain artefacts, for example, foetal heart rate data from non-invasive recordings of the foetal ECG.

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